Conjecture 00000001785: the printed formula is not a step function

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1 Literal necessary conditions

Both versions of the source call the proposed optimal constant a *step function* of c, and give the explicit formula

$$C(c) = F(c) := \max(1, 2c)\,\pi.$$

These two necessary clauses are incompatible on the positive real parameter window $I = (1/2, 1)$. On that window the formula is $F(c) = 2\pi c$, which is injective and has uncountably many values. A step function represented by finitely many interval indicators has only finitely many values. Even a countable interval partition into constant pieces has only countably many values.

We formalize both actual interval representations and their image bounds, rather than defining "step" to mean countable range. The standard finite definition is stated explicitly in Ryzhik's Stanford analysis course [1, Problem 1(i), p. 1]: a finite linear combination of indicators of disjoint intervals. The separate countable-piece result covers a more permissive piecewise-constant reading as well.

This is a disproof of the source's literal necessary conjunction. It does not calculate any actual Ohsawa–Takegoshi optimal constant, assign that constant a value, or refute an independent extension theorem. No convention about the extension's norm or normalization is needed: *no* real function C can satisfy these two clauses on I. Replacing the word "step" by "piecewise linear" would give a different assertion, which this proof does not claim to refute. The word "transition" is not assumed to mean a jump; a change of slope alone would not be a contradiction.

2 A genuine finite interval representation

For a set $A \subseteq \mathbb{R}$, the predicate `FiniteIntervalStepOn A C` requires actual interval sets $J_1, \ldots, J_n$ and real coefficients $a_1, \ldots, a_n$, with

$$J_i \text{ order-connected}, \quad J_i \cap J_j = \varnothing \ (i \neq j), \qquad C(x) = \sum_{i=1}^{n} a_i \mathbf{1}_{J_i}(x) \quad (x \in A).$$

Order-connected subsets of $\mathbb{R}$ are intervals, with any endpoint convention and possibly empty or unbounded intervals. Lean uses the actual set indicator and an actual finite sum.

For each x its membership pattern in the J_i is an element of the finite type $\{0,1\}^n$. Its function value is the sum of the coefficients selected by that pattern. Thus the actual restricted image $C(A)$ is a subset of the finite range of the pattern-value function. The theorem `finite_steps_finite_image` proves this finite-image bridge. It does not postulate the image bound.

3 The countable interval representation

The more permissive predicate `CountableIntervalStepOn A C` requires actual pairwise disjoint order-connected interval sets J_i and coefficients a_i, indexed by $i \in \mathbb{N}$, such that

$$A \subseteq \bigcup_{i \in \mathbb{N}} J_i, \qquad C(x) = a_i \quad (x \in A \cap J_i).$$

This includes countable interval partitions into constant pieces; no local-finiteness assumption is needed for the obstruction. For each $x \in A$, the actual covering condition supplies an index i with $C(x) = a_i$. Consequently $C(A)$ is contained in the actual countable range of $i \mapsto a_i$. Lean proves this conclusion as `countable_steps_countable_image`.

4 The actual formula has uncountable restricted image

For $c \in I$, the strict lower bound $c > 1/2$ gives $2c > 1$, so

$$F(c) = 2c\pi.$$

If $x, y \in I$ and $F(x) = F(y)$, cancellation of the actual nonzero real number π gives $2x = 2y$, hence $x = y$. This proves actual injectivity on the window, using Mathlib's real π.

The real interval I is uncountable. If $F(I)$ were countable, the injective map from the actual interval subtype into its actual image subtype would make I countable. Lean proves this contradiction explicitly. Therefore neither of the two step representations can agree with F throughout I.

5 Quantifiers and the original conjunction

The final necessary-clause theorems quantify over *every* function $C : \mathbb{R} \to \mathbb{R}$ and deny either step representation on I together with $C = F$ there. Additional theorems formally restrict each representation from a larger set to a subset. Since $I \subset (0, \infty)$, Lean also explicitly proves

$$\nexists C : \mathbb{R} \to \mathbb{R} : \quad C \text{ is an interval step function on } (0,\infty) \ \wedge \ \forall c > 0,\ C(c) = \max(1, 2c)\pi,$$

for both the finite and countable representations. Only positive parameters are needed; no claim about negative singular weights or an enlarged parameter domain is used. These incompatible necessary conditions suffice to refute the printed conjecture.

6 Verification

Lean 4.33.0 and Mathlib commit

`db584cd6d46c92f209a44c0f1c829460d327499d`

are pinned. The package includes a new root build, transitive axiom audits of all 15 theorems and full official `leanchecker --fresh Main` replay. No `sorry`, `native_decide` or additional axiom is used. OpenAI Codex assisted; a separate supervising Codex agent reviewed source, report and PDF. This is AI review, not human certification. Organizer acceptance is pending.

References

[1] L. Ryzhik, Stanford Math 205A, Fall 2015, Homework 3 solutions, Problem 1(i), p. 1. `https://math.stanford.edu/~ryzhik/STANFORD/STANF205-16/205A_hw3_sol.pdf`.